

This document provides additional assistance with wiring your Extron IP Link Pro Control Processor to your device. Different components may require a different wiring scheme than those listed below.

For complete operating instructions, refer to the user's manual for the specific IP Link Pro Control Processor or the documentation supplied by the manufacturer of the controlled device.

For more information on using Global Scriptor Modules, refer to the "[Guide to Using Scriptor Modules](#)" document.

Device Specifications

Device Type: Other
Manufacturer: Extron
Firmware Version: N/A
Model(s): DTP HD DA4 4K 230, DTP HD DA4 4K 330, DTP HD DA8 4K 230, DTP HD DA8 4K 330

Tested on the Following Software and Firmware Versions

IP Link Pro Control Processor Firmware	Global Scriptor Version
3.12.0000-b001	2.8.0

Version History

Module Version	Date	Notes
1_2_0_0	12/30/2020	- Updated Ethernet communication to support SSH. - Renamed states in Global HDCP Mode and HDCP Mode. - Polling added to Input HDCP Status and Input Signal Status. - Added Part Number status.
1_1_0_0	10/23/2017	Initial Version

Module Notes

- Unidirectional variable must be set to 'True' if status is not required. Default value is 'False'.
Example: `InterfaceName.Unidirectional = 'True'`
- connectionCounter variable must be set to the number of queries that will be sent to the device before displaying 'Disconnected' if no response is received. Default value is 15.
Example: `InterfaceName.connectionCounter = 5`
- If login credentials are required, devicePassword and deviceUsername must be set accordingly. deviceUsername default value is admin respectively.
Example: `InterfaceName.deviceUsername = 'extron'`

Supported Classes and Examples

SerialClass

```
InterfaceName = ModuleName.SerialClass(ProcessorName, 'COM1', Model='DTP HD DA4 4K 230')
```

SerialOverEthernetClass

```
InterfaceName = ModuleName.SerialOverEthernetClass('192.168.254.254', 2001, Model='DTP HD DA4 4K 230')
```

SSHClass

#Password Required

```
InterfaceName = ModuleName.SSHClass('192.168.254.254', 22023, Credentials=('admin', 'extron'), Model='DTP HD DA4 4K 230')
```

#No Password Required

```
InterfaceName = ModuleName.SSHClass('192.168.254.254', 22023, Credentials=('admin', ''), Model='DTP HD DA4 4K 230')
```

Control Commands

Format with Qualifier:

```
InterfaceName.Set(Command, Value, {'Qualifier Key': 'Qualifier Value'})
```

Format without Qualifier:

```
InterfaceName.Set(Command, Value)
```

Command	Value	Value	
AudioMute	'Off'	'On'	
Qualifier Key	Qualifier Value		
'Output'	'1' – '8' ¹		
# AudioMute example InterfaceName.Set('AudioMute', 'Off', {'Output': '1'})			
Command	Value	Value	
GlobalAudioMute	'On'	'Off'	
# GlobalAudioMute example InterfaceName.Set('GlobalAudioMute', 'On')			
Command	Value	Value	Value
GlobalHDCPMode	'Follow Input'	'Always Encrypt Output'	'Follow Input (With cont DVI trials)'
	'Always Encrypt Output (With cont DVI trials)'		
# GlobalHDCPMode example InterfaceName.Set('GlobalHDCPMode', 'Follow Input')			
Command	Value	Value	
GlobalHDMIAudioMute	'On'	'Off'	
# GlobalHDMIAudioMute example InterfaceName.Set('GlobalHDMIAudioMute', 'On')			
Command	Value	Value	Value
GlobalVideoMute	'On'	'Off'	'Sync'
# GlobalVideoMute example InterfaceName.Set('GlobalVideoMute', 'On')			
Command	Value	Value	
HDCPAuthorization	'Enable'	'Disable'	
# HDCPAuthorization example InterfaceName.Set('HDCPAuthorization', 'Enable')			
Command	Value	Value	Value
HDCPMode	'Follow Input'	'Always Encrypt Output'	'Follow Input (With cont DVI trials)'
	'Always Encrypt Output (With cont DVI trials)'		
Qualifier Key	Qualifier Value		
'Output'	'1' – '8' ¹		
# HDCPMode example InterfaceName.Set('HDCPMode', 'Follow Input', {'Output': '1'})			
Command	Value	Value	
HDMIAudioMute	'Off'	'On'	
Qualifier Key	Qualifier Value		
'Output'	'1' – '8' ¹		
# HDMIAudioMute example InterfaceName.Set('HDMIAudioMute', 'Off', {'Output': '1'})			
Command	Value	Value	Value

Global Scripter Module Communication Sheet

Revision: 12/30/2020

VideoMute	'Off' 'On' 'Sync'
Qualifier Key 'Output'	Qualifier Value '1' – '8' ¹
# VideoMute example InterfaceName.Set('VideoMute', 'Off', {'Output': '1'})	

¹ DTP HD DA4 series has Output states 1 – 4. DTP HD DA8 Series has Output states 1 – 8.

Status Available

For all commands except for InputHDCPStatus, InputSignalStatus, OutputHDCPStatus, and OutputSignalStatus, Update should be called only once since the command's status will be updated automatically as the device's status changes. ConnectionStatus does not support the Update function and is triggered by the device providing a successful response to other Update function calls.

Format with Qualifier:

```
InterfaceName.Update(Command, {'Qualifier Key': 'Qualifier Value'})
Value = InterfaceName.ReadStatus(Command, {'Qualifier Key': 'Qualifier Value'})
InterfaceName.SubscribeStatus(Command, {'Qualifier Key': 'Qualifier Value'},
```

FeedbackHandler)

FeedbackHandler will be called only when the specified qualifier gets a new status.

Format without Qualifier:

```
InterfaceName.Update(Command)
Value = InterfaceName.ReadStatus(Command)
InterfaceName.SubscribeStatus(Command, None, FeedbackHandler)
FeedbackHandler will be called when any qualifier gets a new status.
```

Command	Value	Value	
AudioMute	'Off'	'On'	
Qualifier Key	Qualifier Value		
'Output'	'1' – '8' ¹		
# AudioMute example InterfaceName.Update('AudioMute', {'Output': '1'}) Value = InterfaceName.ReadStatus('AudioMute', {'Output': '1'}) InterfaceName.SubscribeStatus('AudioMute', None, FeedbackHandler)			
Command	Value	Value	
ConnectionStatus	'Connected'	'Disconnected'	
# ConnectionStatus example Value = InterfaceName.ReadStatus('ConnectionStatus') InterfaceName.SubscribeStatus('ConnectionStatus', None, FeedbackHandler)			
Command	Value	Value	
HDCPAuthorization	'Enable'	'Disable'	
# HDCPAuthorization example InterfaceName.Update('HDCPAuthorization') Value = InterfaceName.ReadStatus('HDCPAuthorization') InterfaceName.SubscribeStatus('HDCPAuthorization', None, FeedbackHandler)			
Command	Value	Value	Value
HDCPMode	'Follow Input'	'Always Encrypt Output'	'Follow Input (With cont DVI trials)'
	'Always Encrypt Output (With cont DVI trials)'		
Qualifier Key	Qualifier Value		
'Output'	'1' – '8' ¹		
# HDCPMode example InterfaceName.Update('HDCPMode', {'Output': '1'}) Value = InterfaceName.ReadStatus('HDCPMode', {'Output': '1'}) InterfaceName.SubscribeStatus('HDCPMode', None, FeedbackHandler)			
Command	Value	Value	
HDMIAudioMute	'Off'	'On'	
Qualifier Key	Qualifier Value		
'Output'	'1' – '8' ¹		
# HDMIAudioMute example InterfaceName.Update('HDMIAudioMute', {'Output': '1'})			

Global Scripter Module Communication Sheet

Revision: 12/30/2020

Value = InterfaceName.ReadStatus('HDMIAudioMute', {'Output': '1'}) InterfaceName.SubscribeStatus('HDMIAudioMute', None, FeedbackHandler)			
Command	Value		Value
InputHDCPStatus	'Enabled'		'Disabled'
Qualifier Key	Qualifier Value		Qualifier Value
'Type'	'Loop Through'		'HDMI'
# InputHDCPStatus example InterfaceName.Update('InputHDCPStatus', {'Type': 'Loop Through'}) Value = InterfaceName.ReadStatus('InputHDCPStatus', {'Type': 'Loop Through'}) InterfaceName.SubscribeStatus('InputHDCPStatus', None, FeedbackHandler)			
Command	Value		Value
InputSignalStatus	'Enabled'		'Disabled'
Qualifier Key	Qualifier Value		Qualifier Value
'Type'	'Loop Through'		'HDMI'
# InputSignalStatus example InterfaceName.Update('InputSignalStatus', {'Type': 'Loop Through'}) Value = InterfaceName.ReadStatus('InputSignalStatus', {'Type': 'Loop Through'}) InterfaceName.SubscribeStatus('InputSignalStatus', None, FeedbackHandler)			
Command	Value		Value
OutputHDCPStatus	'Enabled'		'Disabled'
Qualifier Key	Qualifier Value		
'Output'	'1' – '8' ¹		
# OutputHDCPStatus example InterfaceName.Update('OutputHDCPStatus', {'Output': '1'}) Value = InterfaceName.ReadStatus('OutputHDCPStatus', {'Output': '1'}) InterfaceName.SubscribeStatus('OutputHDCPStatus', None, FeedbackHandler)			
Command	Value		Value
OutputSignalStatus	'Enabled'		'Disabled'
Qualifier Key	Qualifier Value		
'Output'	'1' – '8' ¹		
# OutputSignalStatus example InterfaceName.Update('OutputSignalStatus', {'Output': '1'}) Value = InterfaceName.ReadStatus('OutputSignalStatus', {'Output': '1'}) InterfaceName.SubscribeStatus('OutputSignalStatus', None, FeedbackHandler)			
Command	Value		
PartNumber	'String'		
# PartNumber example InterfaceName.Update('PartNumber') Value = InterfaceName.ReadStatus('PartNumber') InterfaceName.SubscribeStatus('PartNumber', None, FeedbackHandler)			
Command	Value	Value	Value
VideoMute	'Off'	'On'	'Sync'
Qualifier Key	Qualifier Value		
'Output'	'1' – '8' ¹		
# VideoMute example InterfaceName.Update('VideoMute', {'Output': '1'}) Value = InterfaceName.ReadStatus('VideoMute', {'Output': '1'}) InterfaceName.SubscribeStatus('VideoMute', None, FeedbackHandler)			

¹ DTP HD DA4 series has Output states 1 – 4. DTP HD DA8 Series has Output states 1 – 8.

Cable and Adapter Requirements

Captive Screw to Captive Screw RS232 Serial Cable

Notes for the Device

Serial communication

Port Type: RS-232, Serial_RS422

Baud Rate: 9600

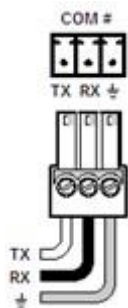
Data Bits: 8

Parity: None

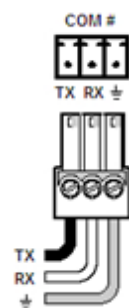
Stop Bits: One

Flow Control: None

Pin Assignments Diagram



Signal	Main Cable	Signal
TxD		TxD
RxD		RxD
GND		GND



Network communication

When configuring the Ethernet module, be sure device settings match those of the Global Scriptor ethernet interface.

Port Type: Ethernet

Default Port: 22023

Logon Credentials Supported: Yes

Default Username: admin

Default Password:

Multi-Connection Yes

Capabilities:

Port Changeability: No

Ethernet Module Configuration Description

Please refer to user manual for settings and changes to the network communication

Notes for the Device

Appendix A. Set Commands

Audio Mute Off Output 1	1*0Z
Audio Mute Off Output 4	4*0Z
Audio Mute Off Output 8	8*0Z
Audio Mute On Output 1	1*1Z
Audio Mute On Output 4	4*1Z
Audio Mute On Output 8	8*1Z
Global Audio Mute Off	0Z\x0D
Global Audio Mute On	1Z\x0D
Global HDCP Mode Always Encrypt Output	WS1HDCP\x0D
Global HDCP Mode Always Encrypt Output (With cont DVI trials)	WS3HDCP\x0D
Global HDCP Mode Follow Input	WS0HDCP\x0D
Global HDCP Mode Follow Input (With cont DVI trials)	WS2HDCP\x0D
Global HDMI Audio Mute Off	W0AFMT\x0D
Global HDMI Audio Mute On	W1AFMT\x0D
Global Video Mute Off	0B\x0D
Global Video Mute On	1B\x0D
Global Video Mute Sync	2B\x0D
HDCP Authorization Disable	WE0HDCP\x0D
HDCP Authorization Enable	WE1HDCP\x0D
HDCP Mode Always Encrypt Output (With cont DVI trials) Output 1	WS1*3HDCP\x0D
HDCP Mode Always Encrypt Output (With cont DVI trials) Output 4	WS4*3HDCP\x0D
HDCP Mode Always Encrypt Output (With cont DVI trials) Output 8	WS8*3HDCP\x0D
HDCP Mode Always Encrypt Output Output 1	WS1*1HDCP\x0D
HDCP Mode Always Encrypt Output Output 4	WS4*1HDCP\x0D
HDCP Mode Always Encrypt Output Output 8	WS8*1HDCP\x0D
HDCP Mode Follow Input (With cont DVI trials) Output 1	WS1*2HDCP\x0D
HDCP Mode Follow Input (With cont DVI trials) Output 4	WS4*2HDCP\x0D
HDCP Mode Follow Input (With cont DVI trials) Output 8	WS8*2HDCP\x0D
HDCP Mode Follow Input Output 1	WS1*0HDCP\x0D
HDCP Mode Follow Input Output 4	WS4*0HDCP\x0D
HDCP Mode Follow Input Output 8	WS8*0HDCP\x0D
HDMI Audio Mute Off Output 1	W1*0AFMT\x0D
HDMI Audio Mute Off Output 4	W4*0AFMT\x0D
HDMI Audio Mute Off Output 8	W8*0AFMT\x0D
HDMI Audio Mute On Output 1	W1*1AFMT\x0D

Global Scriptor Module Communication Sheet

Revision: 12/30/2020

HDMI Audio Mute On Output 4	W4*1AFMT\x0D
HDMI Audio Mute On Output 8	W8*1AFMT\x0D
Video Mute Off Output 1	1*0B
Video Mute Off Output 4	4*0B
Video Mute Off Output 8	8*0B
Video Mute On Output 1	1*1B
Video Mute On Output 4	4*1B
Video Mute On Output 8	8*1B
Video Mute Sync Output 1	1*2B
Video Mute Sync Output 4	4*2B
Video Mute Sync Output 8	8*2B

Appendix B. Update Commands

Audio Mute Output 1	Z
Audio Mute Output 4	Z
Audio Mute Output 8	Z
HDCP Authorization	WEHDCP\x0D
HDCP Mode Output 1	WSHDCP\x0D
HDCP Mode Output 4	WSHDCP\x0D
HDCP Mode Output 8	WSHDCP\x0D
HDMI Audio Mute Output 1	WAFMT\x0D
HDMI Audio Mute Output 4	WAFMT\x0D
HDMI Audio Mute Output 8	WAFMT\x0D
Output HDCP Status Output 1	WHDCP\x0D
Output HDCP Status Output 4	WHDCP\x0D
Output HDCP Status Output 8	WHDCP\x0D
Output Signal Status Output 1	WLS\x0D
Output Signal Status Output 4	WLS\x0D
Output Signal Status Output 8	WLS\x0D
Part Number	N
Video Mute Output 1	B
Video Mute Output 4	B
Video Mute Output 8	B